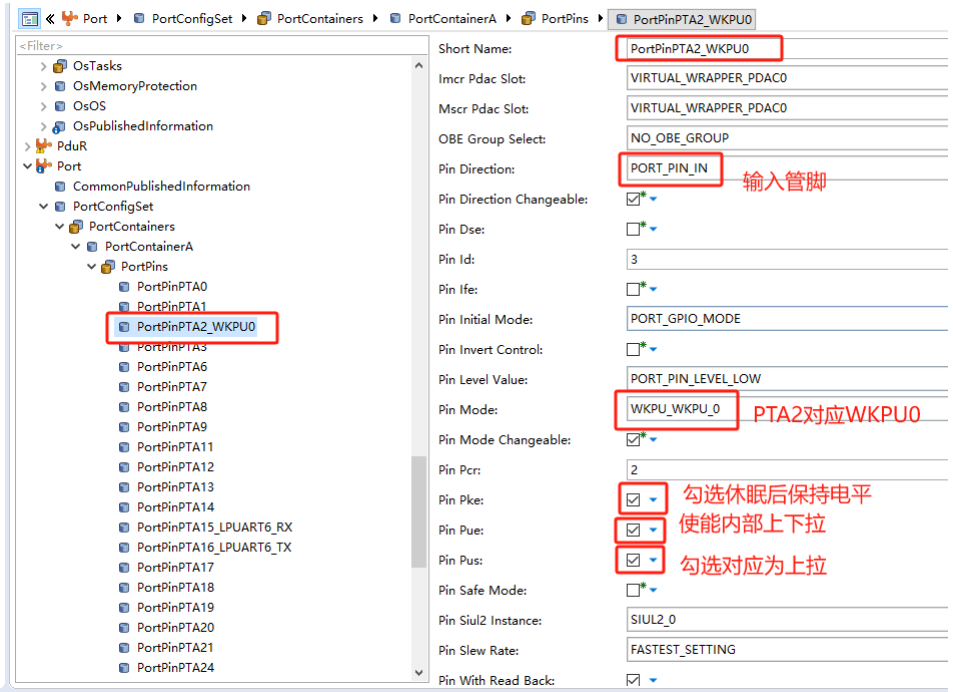
23、Icu模块的Wkpu中断边沿

卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

一、修改Port PTA2

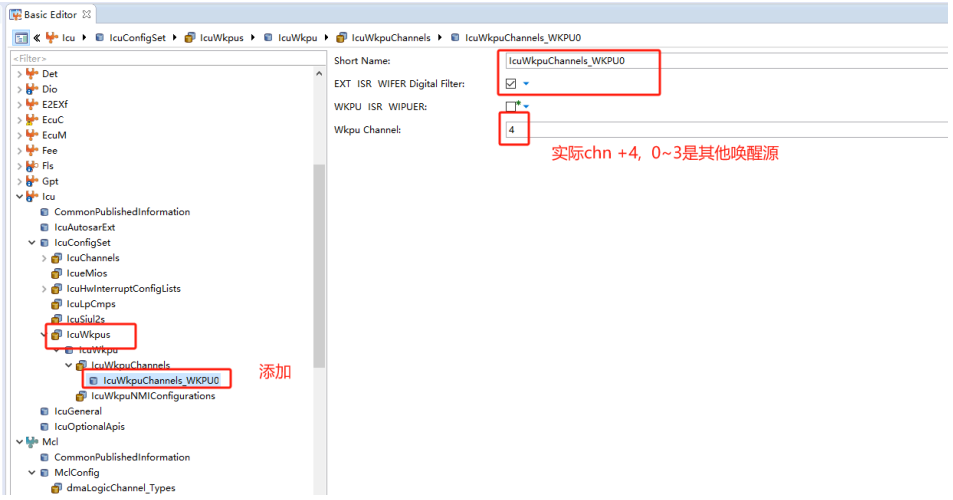
这里设置内部上拉，所以我们触发方式就是下降沿，接地即可触发



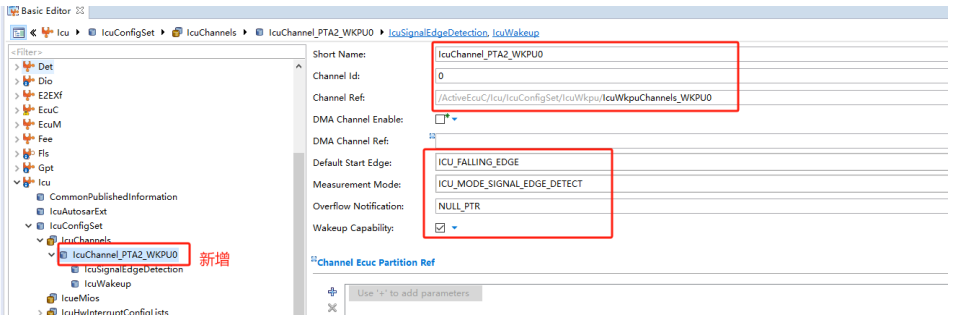
二、添加Icu、Mcl模块

Mcl 模块不用修改什么，目前只是为了编译通过

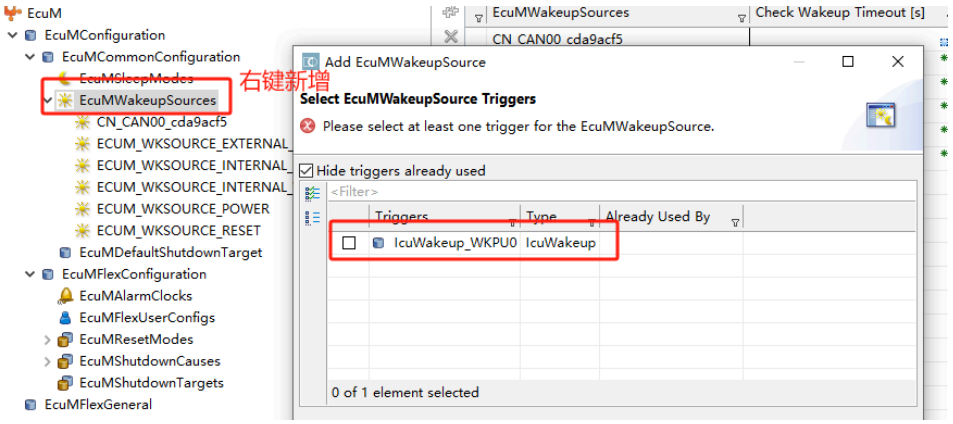
Icu 添加



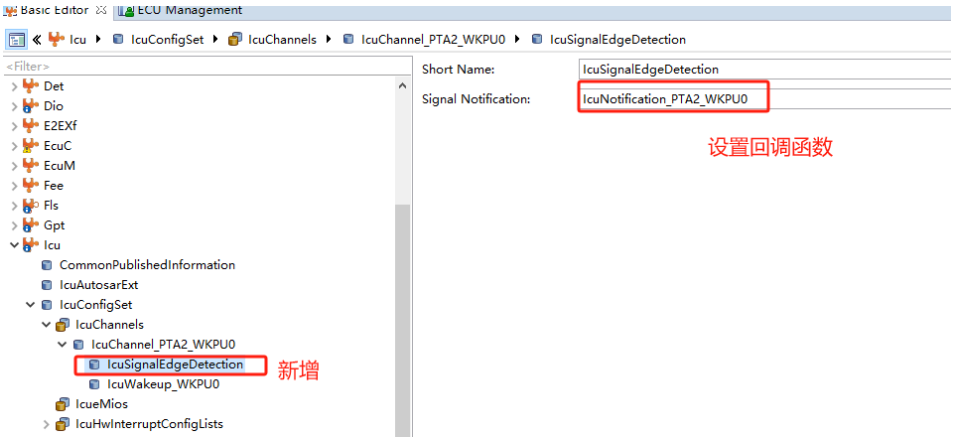
设置EDGE_DETECT，下降沿

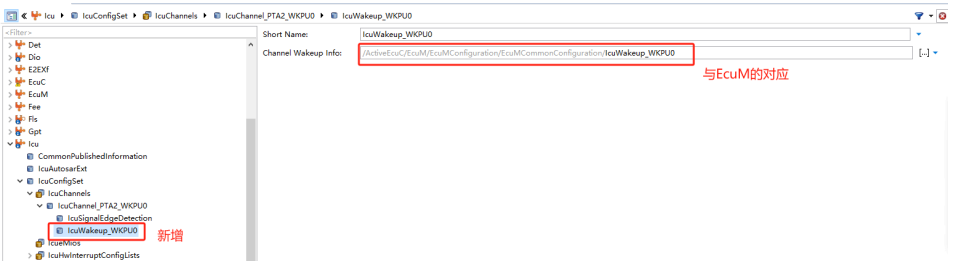


EcuM添加WakpSource

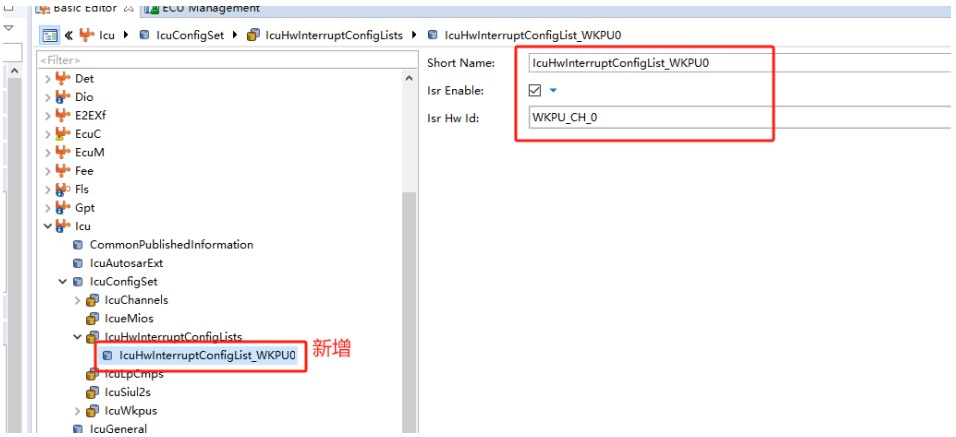


Icu



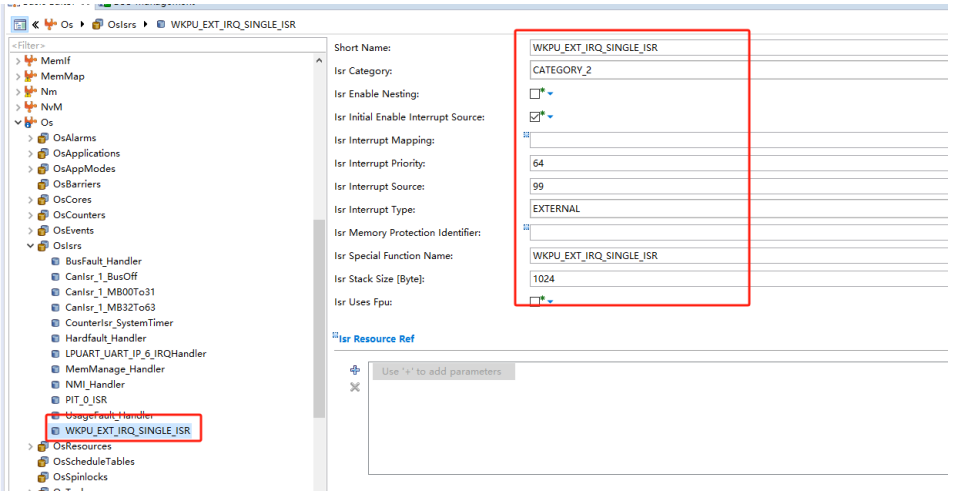


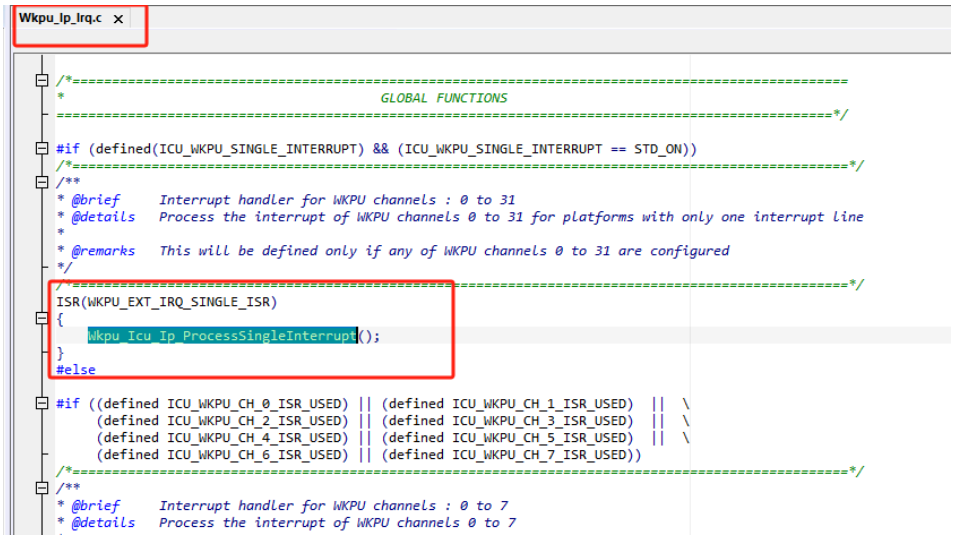
添加中断配置，使能，这里很重要，不配置就不会使能对应中断的宏



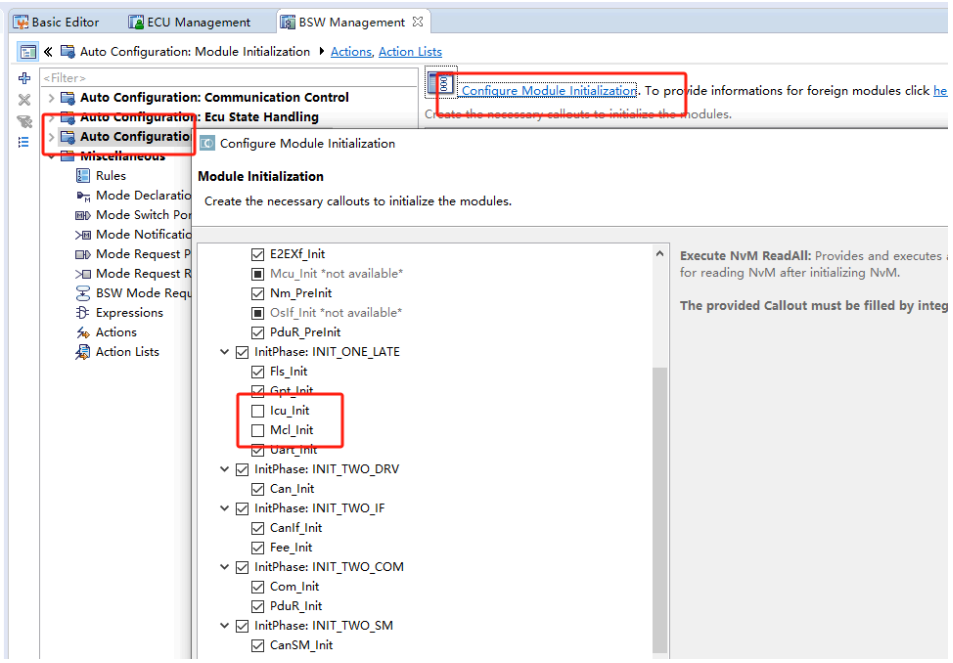
三、Os

添加中断，中断函数名在mcal的icu里面找到





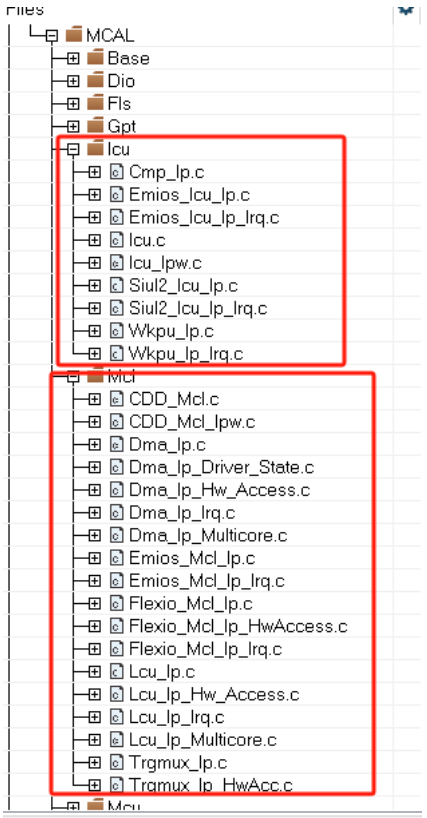
四、BswM添加Init



生成全部模块

五、IAR添加相关文件编译

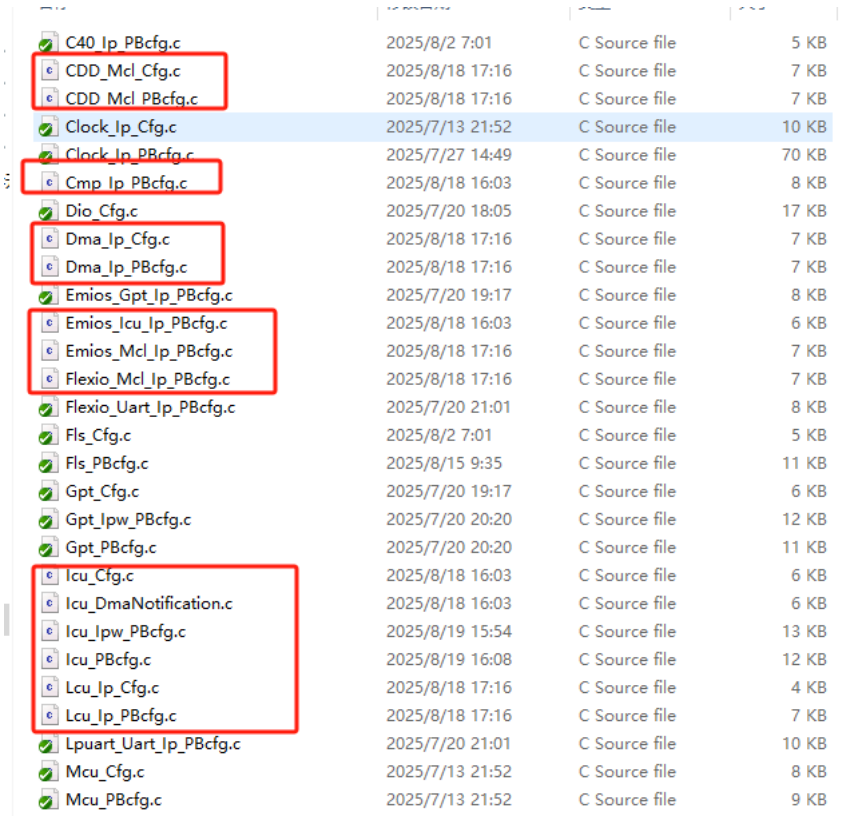
5.1、添加MCAL的icu和mcl，MCAL下新建Icu组和Mcl组，并添加mcal中对应的文件



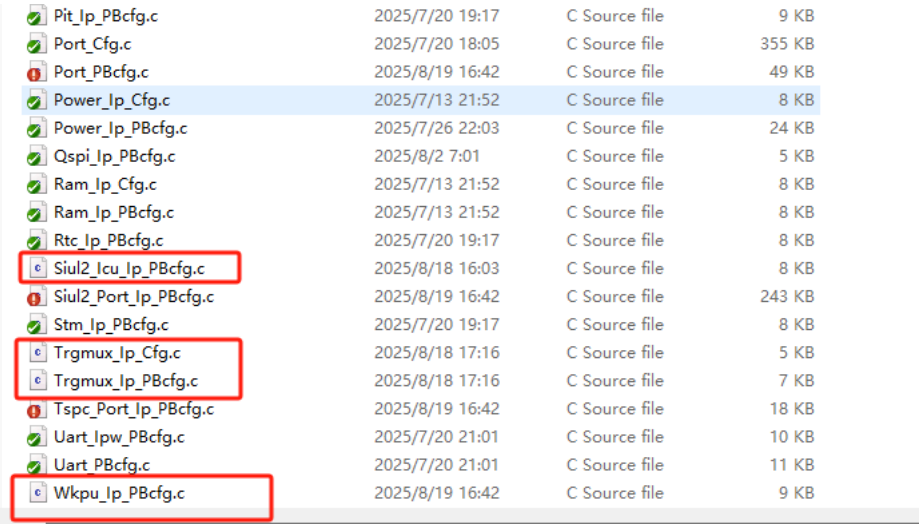
添加mcal的头文件路径到IAR

$PROJ_DIR$\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Icu_TS_T40D34M
$PROJ_DIR$\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Mcl_TS_T40D34

5.2、添加davinci生成的icu/mcl文件



C40_Ip_PBcfg.c	2025/8/2 7:01	C Source file	5 KB
CDD_Mcl_Cfg.c	2025/8/18 17:16	C Source file	7 KB
CDD_Mcl_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Clock_Ip_Cfg.c	2025/7/13 21:52	C Source file	10 KB
Clock_Ip_PBcfg.c	2025/7/27 14:49	C Source file	70 KB
Cmp_Ip_PBcfg.c	2025/8/18 16:03	C Source file	8 KB
Dio_Cfg.c	2025/7/20 18:05	C Source file	17 KB
Dma_Ip_Cfg.c	2025/8/18 17:16	C Source file	7 KB
Dma_Ip_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Emios_Gpt_Ip_PBcfg.c	2025/7/20 19:17	C Source file	8 KB
Emios_Icu_Ip_PBcfg.c	2025/8/18 16:03	C Source file	6 KB
Emios_Mcl_Ip_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Flexio_Mcl_Ip_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Flexio_Uart_Ip_PBcfg.c	2025/7/20 21:01	C Source file	8 KB
Fls_Cfg.c	2025/8/2 7:01	C Source file	5 KB
Fls_PBcfg.c	2025/8/15 9:35	C Source file	11 KB
Gpt_Cfg.c	2025/7/20 19:17	C Source file	6 KB
Gpt_Ipw_PBcfg.c	2025/7/20 20:20	C Source file	12 KB
Gpt_PBcfg.c	2025/7/20 20:20	C Source file	11 KB
Icu_Cfg.c	2025/8/18 16:03	C Source file	6 KB
Icu_DmaNotification.c	2025/8/18 16:03	C Source file	6 KB
Icu_Ipw_PBcfg.c	2025/8/19 15:54	C Source file	13 KB
Icu_PBcfg.c	2025/8/19 16:08	C Source file	12 KB
Lcu_Ip_Cfg.c	2025/8/18 17:16	C Source file	4 KB
Lcu_Ip_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Lpuart_Uart_Ip_PBcfg.c	2025/7/20 21:01	C Source file	10 KB
Mcu_Cfg.c	2025/7/13 21:52	C Source file	8 KB
Mcu_PBcfg.c	2025/7/13 21:52	C Source file	9 KB



Pit_Ip_PBcfg.c	2025/7/20 19:17	C Source file	9 KB
Port_Cfg.c	2025/7/20 18:05	C Source file	355 KB
Port_PBcfg.c	2025/8/19 16:42	C Source file	49 KB
Power_Ip_Cfg.c	2025/7/13 21:52	C Source file	8 KB
Power_Ip_PBcfg.c	2025/7/26 22:03	C Source file	24 KB
Qspi_Ip_PBcfg.c	2025/8/2 7:01	C Source file	5 KB
Ram_Ip_Cfg.c	2025/7/13 21:52	C Source file	8 KB
Ram_Ip_PBcfg.c	2025/7/13 21:52	C Source file	8 KB
Rtc_Ip_PBcfg.c	2025/7/20 19:17	C Source file	8 KB
Siul2_Icu_Ip_PBcfg.c	2025/8/18 16:03	C Source file	8 KB
Siul2_Port_Ip_PBcfg.c	2025/8/19 16:42	C Source file	243 KB
Stm_Ip_PBcfg.c	2025/7/20 19:17	C Source file	8 KB
Trgmux_Ip_Cfg.c	2025/8/18 17:16	C Source file	5 KB
Trgmux_Ip_PBcfg.c	2025/8/18 17:16	C Source file	7 KB
Tspc_Port_Ip_PBcfg.c	2025/8/19 16:42	C Source file	18 KB
Uart_Ipw_PBcfg.c	2025/7/20 21:01	C Source file	10 KB
Uart_PBcfg.c	2025/7/20 21:01	C Source file	11 KB
Wkpu_Ip_PBcfg.c	2025/8/19 16:42	C Source file	9 KB

5.3、编译报错修改

```
Messages
Warning[Pa181]: incompatible redefinition of macro "VAR_ALIGN" (defined at line 161)
Warning[Pe161]: unrecognized #pragma
Warning[Pe161]: unrecognized #pragma
Fatal Error[Pe1696]: cannot open source file "EcuM_Externals.h"
        searched: "D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Icu_TS_T40D34M20I1R0\include\"
        searched: "D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\DavinciProject\Appl\GenData\src\"
```

修改Icu.h

```
...\CBD2300384_D00\ThirdParty\Mcal_S32k\Supply\SW32K3_RTD_4.4_2.0.1\eclipse\plugins\Icu_TS_T40D34M20I1R0\include\Icu.h
)25/8/19 18:05:04   47,081 字节   C,C++,C#,ObjC 源代码   UTF-8   PC

#ifdef __cplusplus
extern "C"
{
#endif

/*==================================================================================================
*                                          INCLUDE FILES
*  1) system and project includes
*  2) needed interfaces from external units
*  3) internal and external interfaces from this unit
==================================================================================================*/
#include "Mcal.h"
#include "StandardTypes.h"
#include "Icu_Cfg.h"
#include "Icu_Irq.h"
#include "Icu_Types.h"
#include "Icu_Ipw_Types.h"

#if (ICU_REPORT_WAKEUP_SOURCE == STD_ON)
//#include "EcuM_Externals.h"
#include "EcuM_Cbk.h"
#endif
```

另外mcal包中Base模块include下 _Icu_MemMap.h和_Mcl_MemMap.h 都把前面的 _ 去掉

```
CBD2300384_D00/ThirdParty/Mcal_S32k/Supply/SW32K3_RTD_4.4_2.0.1/eclipse/plugins/Base_TS_T40D34M20I1R0/include/Icu_MemMap.h   .h   未知
CBD2300384_D00/ThirdParty/Mcal_S32k/Supply/SW32K3_RTD_4.4_2.0.1/eclipse/plugins/Base_TS_T40D34M20I1R0/include/Mcl_MemMap.h   .h   未知
```

5.4、添加测试代码

启动回调、唤醒、边缘检测

```
CtApTestSwc.c | CtApDiagSwc.c | CtApCanApp.c | CtApNvMSwc.c | ARMStartup_CortexM.c | Can.c | Com_TxModeHdlr.c | Com_SignalIf.c | PduR

    Dio_FlipChannel(DioConf_DioChannel_DioChannelPTA29_LED1_RED);
  }

  /**********************************************************************************************
   * DO NOT CHANGE THIS COMMENT!          << End of documentation area >>                DO NOT CHANGE
   **********************************************************************************************

  FUNC(void, CtApTestSwc_CODE) RCtApTestSwc_Init(void) /* PRQA S 0624, 3206 */ /* MD_Rte_0624, MD_Rte_3206
  {
  /**********************************************************************************************
   * DO NOT CHANGE THIS COMMENT!          << Start of runnable implementation >>         DO NOT CHANGE
   * Symbol: RCtApTestSwc_Init
   **********************************************************************************************
    printf("Swc Init\r\n");
    Rte_Write_Request_ESH_RunRequest_0_requestedMode(REQUESTED);
    Rte_Write_Request_ESH_RunRequest_1_requestedMode(REQUESTED);
    Gpt_EnableNotification(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0);
    Gpt_StartTimer(GptConf_GptChannelConfiguration_GptChannelConfiguration_PIT0_CH0, 40000000);

    Icu_EnableNotification(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);
    Icu_EnableWakeup(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);
    Icu_EnableEdgeDetection(IcuConf_IcuChannel_IcuChannel_PTA2_WKPU0);
   **********************************************************************************************
   * DO NOT CHANGE THIS COMMENT!          << End of runnable implementation >>           DO NOT CHANGE
   **********************************************************************************************
  }
```

添加中断回调函数，当跑起来后，PTA2 从悬空到接地，就会进入这个中断回调函数，中断回调函数名是跟icu模块里面设置的一样的

```
CtApTestSwc.c × | CtApDiagSwc.c | CtApCanApp.c | CtApNvMSwc.c | ARMStartup_CortexM.c | Can.c | Co

/*****************************************************************************
 *
 * Runnable Entity Name: RCtApTestSwc_Init
 *
 *----------------------------------------------------------------------------
 *
 * Executed once after the RTE is started
 *
 *****************************************************************************
/*****************************************************************************
 * DO NOT CHANGE THIS COMMENT!           << Start of documentation area >>
 * Symbol: RCtApTestSwc_Init_doc
 *****************************************************************************
void IcuNotification_PTA2_WKPU0(void)
{

}

void GptNotification_PIT0_CH0(void)
{
  Dio_FlipChannel(DioConf_DioChannel_DioChannelPTA29_LED1_RED);
}
```